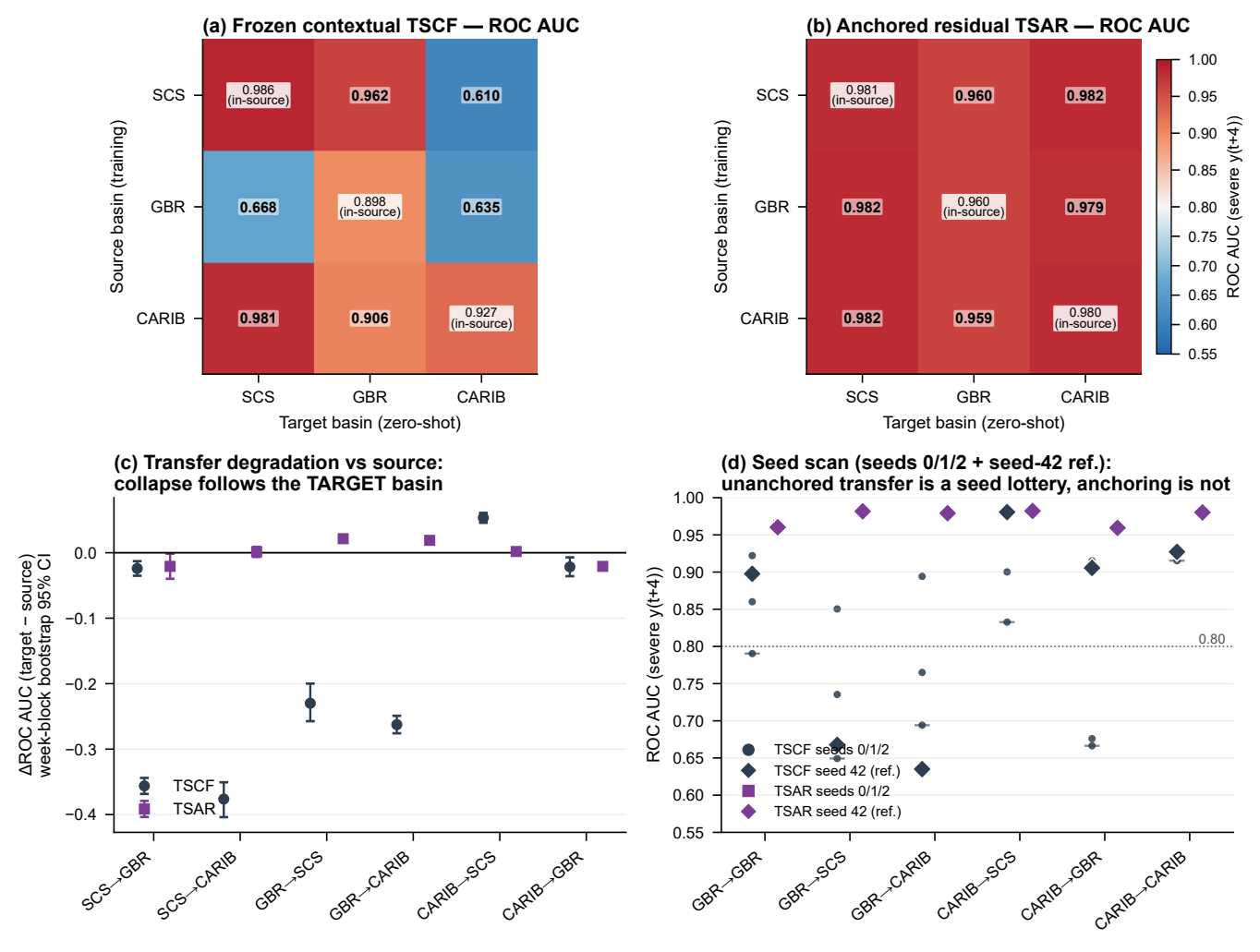

(a) Frozen contextual TSCF — ROC AUC
Source basin (training)
SCS
GBR
CARIB
0.986 (in-source)
0.962
0.610
0.668
0.898 (in-source)
0.635
0.981
0.906
0.927 (in-source)
SCS
GBR
CARIB
Target basin (zero-shot)
(b) Anchored residual TSAR — ROC AUC
Source basin (training)
SCS
GBR
CARIB
0.981 (in-source)
0.960
0.982
0.982
0.960 (in-source)
0.979
0.982
0.959
0.980 (in-source)
SCS
GBR
CARIB
Target basin (zero-shot)
ROC AUC (severe y(t+4))
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
(c) Transfer degradation vs source:
collapse follows the TARGET basin
ΔROC AUC (target − source)
week-block bootstrap 95% CI
0.0
−0.1
−0.2
−0.3
−0.4
TSCF
TSAR
SCS→GBR
SCS→CARIB
GBR→SCS
GBR→CARIB
CARIB→SCS
CARIB→GBR
(d) Seed scan (seeds 0/1/2 + seed-42 ref.):
unanchored transfer is a seed lottery, anchoring is not
ROC AUC (severe y(t+4))
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.80
TSCF seeds 0/1/2
TSCF seed 42 (ref.)
TSAR seeds 0/1/2
TSAR seed 42 (ref.)
GBR→GBR
GBR→SCS
GBR→CARIB
CARIB→SCS
CARIB→GBR
CARIB→CARIB